

Podcast 29 — AI/ML Services

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Length target: 15 min **Repo references:** `domain-2-new-solutions/35-bedrock.md` , `36-sagemaker.md` , `48-ml-services.md`

Topic & Scope

Bedrock (generative AI / foundation models), SageMaker (custom ML platform), plus the pre-trained AI service bundle (Comprehend, Polly, Transcribe, Rekognition, Textract, Translate, Lex, Kendra, Personalize, Fraud Detector). SAP-C02 mostly tests “which service for which input/output”.

Service Coverage Depth

Deep: - Bedrock — foundation models, Knowledge Bases (RAG), Agents, Guardrails, Provisioned Throughput - SageMaker — training, inference (real-time / async / batch / serverless), endpoints, multi-model + multi-container

Brief — “input/output recognition”: - Comprehend (text NLP) - Polly (text-to-speech) - Transcribe (speech-to-text) - Rekognition (image/video) - Textract (OCR + forms) - Translate (translation) - Lex (chatbots) - Kendra (enterprise search) - Personalize (recommendations) - Fraud Detector

Structured Outline

1. **Open (30s)** — “Generative AI is on the exam. Plus SageMaker and the pre-trained AI bundle. Memorize: which input becomes which output.”
2. **Bedrock (5 min)** — foundation models from Anthropic / Meta / Mistral / AI21 / Cohere / Amazon Titan / Stability, Knowledge Bases (RAG with vector DB), Agents (tool use), Guardrails (content filtering), Provisioned Throughput, Model evaluation
3. **SageMaker (4 min)** — Studio (IDE), training (managed compute), inference modes (real-time, async, batch, serverless), endpoint deployment (single model, multi-model, multi-container, inference pipeline), Feature Store, Model Monitor, Clarify (bias detection), Pipelines (ML CI/CD), Ground Truth (labeling), JumpStart (model zoo)
4. **Pre-trained AI services Quick Round (4 min)** — verbalize each as “input → output”:
Comprehend (text→sentiment/entities), Polly (text→speech), Transcribe (speech→text), Rekognition (image/video→labels), Textract (document→structured fields), Translate (text→translated text), Lex (utterance→bot response), Kendra (query→ranked documents), Personalize (interactions→recommendations), Fraud Detector (transaction→risk score)
5. **Decision Matrix (1 min)**
6. **Rapid-Fire Trap Drill (60s)**

Must-Mention Exam Tips

- Bedrock: managed access to foundation models via API; no infrastructure
- Bedrock Knowledge Bases: managed RAG with vector embeddings (Aurora pgvector / OpenSearch / Pinecone)
- Bedrock Agents: tools + reasoning loop; calls Lambda or APIs
- Bedrock Guardrails: content policy + PII redaction + topic blocking + grounding
- Provisioned Throughput: dedicated capacity for production at lower per-token cost
- SageMaker training: built-in algorithms, BYO container, Spot training for cost
- SageMaker inference: real-time (low latency, always on), async (queue, large payloads, GPU), batch (S3 in/out), serverless (cold-start tolerant)
- SageMaker Multi-Model Endpoint: many models on one endpoint, lazy-load
- SageMaker Feature Store: online + offline feature serving
- SageMaker Pipelines: ML CI/CD with model registry + approval workflow
- SageMaker Clarify: bias detection + explainability
- SageMaker Model Monitor: data drift detection
- SageMaker Ground Truth: managed labeling workflow
- SageMaker JumpStart: pre-trained models, one-click deploy
- Comprehend: sentiment, entities, key phrases, classification (custom too)
- Comprehend Medical: HIPAA medical text
- Polly: TTS, 60+ voices, SSML, neural voices
- Transcribe: ASR, real-time + batch, speaker ID, custom vocab, PII redaction
- Rekognition: image/video labels, faces, content moderation, custom labels
- Textract: OCR + forms + tables, document Q&A queries
- Translate: 75+ languages, custom terminology, document translation preserves format
- Lex: chatbots — same engine as Alexa
- Kendra: enterprise search with NLP
- Personalize: recommendation engine
- Fraud Detector: ML-based fraud risk scoring

Must-Mention Exam Traps

- EXAM TRAP: Bedrock is for FOUNDATION MODELS — not for training your own from scratch (use SageMaker)
- EXAM TRAP: Bedrock data is NOT used to train base models — your prompts/responses stay private
- EXAM TRAP: SageMaker endpoints have cold starts on serverless mode — for real-time low-latency use real-time endpoint
- EXAM TRAP: SageMaker async inference for large payloads + variable latency — uses queue + SNS notification

- EXAM TRAP: Comprehend ≠ Texttract — Comprehend analyzes text; Texttract extracts structured data from documents
- EXAM TRAP: Lex builds bots; for free-form generation use Bedrock
- EXAM TRAP: Kendra is search — for RAG pair with Bedrock
- EXAM TRAP: Personalize needs interaction history — won't work cold-start
- EXAM TRAP: Rekognition Custom Labels ≠ SageMaker — Custom Labels is no-code; SageMaker is full ML platform
- EXAM TRAP: SageMaker Studio Notebooks are billed when running — stop when not in use
- EXAM TRAP: Bedrock Knowledge Base uses vector DB — OpenSearch Serverless is the default cheap option
- EXAM TRAP: SageMaker multi-model endpoint requires models in S3 + container that supports multi-model
- EXAM TRAP: Some AI services are not in every Region — check before designing

Key Decision Matrix

Input → Output	Pick
Text → sentiment / entities	Comprehend
Text → speech	Polly
Speech → text transcript	Transcribe
Image / video → objects / labels	Rekognition
Document → structured form data	Texttract
Text → translated text	Translate
User → chatbot reply	Lex
Query → ranked enterprise docs	Kendra
User events → recommendation	Personalize
Transaction → fraud score	Fraud Detector
Prompt → generated text / code / image	Bedrock
Custom ML model training	SageMaker
Generative AI with grounded enterprise docs	Bedrock + Knowledge Base
LLM agent with tool calls	Bedrock Agents
Block toxic / off-topic LLM output	Bedrock Guardrails

Tone & Style

- For each pre-trained AI service, give the “input → output” sentence and one use case
- Bedrock + SageMaker get the deep treatment; everything else is recognition
- Repeat “Bedrock for foundation models, SageMaker for your own”

Rapid-Fire Closer

“OCR + forms?” — “Textract.” “Text → speech?” — “Polly.” “Speech → text?” — “Transcribe.”
“Chatbot?” — “Lex.” “Enterprise search?” — “Kendra.” “Recommendations?” — “Personalize.”
“Foundation models?” — “Bedrock.” “Custom ML?” — “SageMaker.”